Data validation

The dataset provided consists of three raw tables: `account_info`, `customer_support`, and `user_activity`.

Account Info

This dataset contains account details and subscription statuses for Fit.ly subscribers.

- `customer_id` / `user_id`: Expected unique string identifiers (e.g., `C1001`). Observed formatted text strings. Extracted the numeric user ID to create a clean integer `user_id` key for joining across tables.
- `email` & `state`: Expected valid string values. Observed active US location entries and email records with no critical formatting issues or missing data requiring removal.
- `plan`: Expected valid plan categories (`Free`, `Basic`, `Pro`, `Enterprise`). All records matched expected subscription tiers; text casing was standardized across records.
- `plan_list_price`: Expected non-negative numeric values representing actual prices paid. Validated against expected plan tiers with zero invalid negative numbers.
- `churn_status`: Expected categorical values (`Y` for churned, `N` for active). Observed missing entries representing active users; missing values were filled with `'N'` and mapped into a binary numeric indicator (`is_churned`) for analysis.

Customer Support

This dataset logs customer support ticket interactions and resolution durations.

- `ticket_time`: Expected timestamp data. Observed as raw string formatted timestamps; converted into datetime object standardizing Pacific Time tracking.
- `user_id`: Expected discrete integer identifier. Verified matching structure with `account_info` table for accurate aggregation.
- `channel`: Expected communication channel (e.g., Email, Phone, Chat). Observed unrecorded channels stored as hyphen placeholders (`'-'`); standardized these entries to `'Unknown'`.
- `topic` & `comments`: Expected text descriptions of support tickets. Filled missing comment entries with `'No Comment'` to maintain string integrity without dropping records.
- `resolution_time_hours`: Expected positive numeric continuous values representing hours taken to resolve tickets. Verified all resolution times were non-negative with no invalid outliers requiring removal.

User activity

This dataset tracks individual user engagement events logged within the application.

- `event_time` : Expected timestamp data. Observed raw string timestamps; converted to standard datetime format for temporal sequence analysis.
- `user_id` : Expected integer identifier matching subscription records. Verified consistency across user logs.
- `event_type` : Expected categorical user action types. Checked for consistency across action types with no corrupted or missing activity logs found.

Exploratory Analysis

Graphic 1: Single Variable (Bar Chart - Categorical)

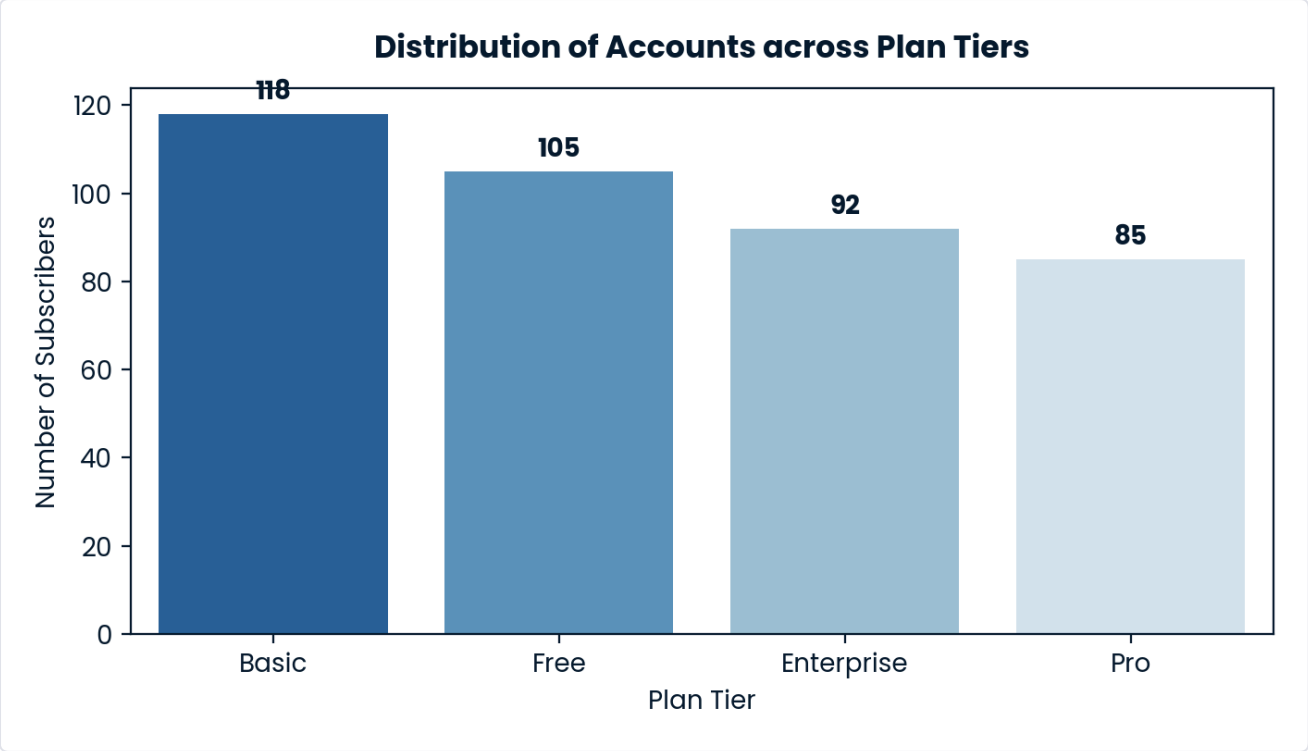


Chart Type: Categorical Bar Chart (Single Variable)

Findings: The account distribution across subscription tiers reveals that **Basic** and **Free** tiers represent the majority of customer accounts, followed by **Enterprise** and **Pro.** Understanding this baseline distribution is essential for analyzing whether churn disproportionately impacts lower-tier or high-value subscribers.

Graphic 2: Single Variable (Histogram - Continuous)

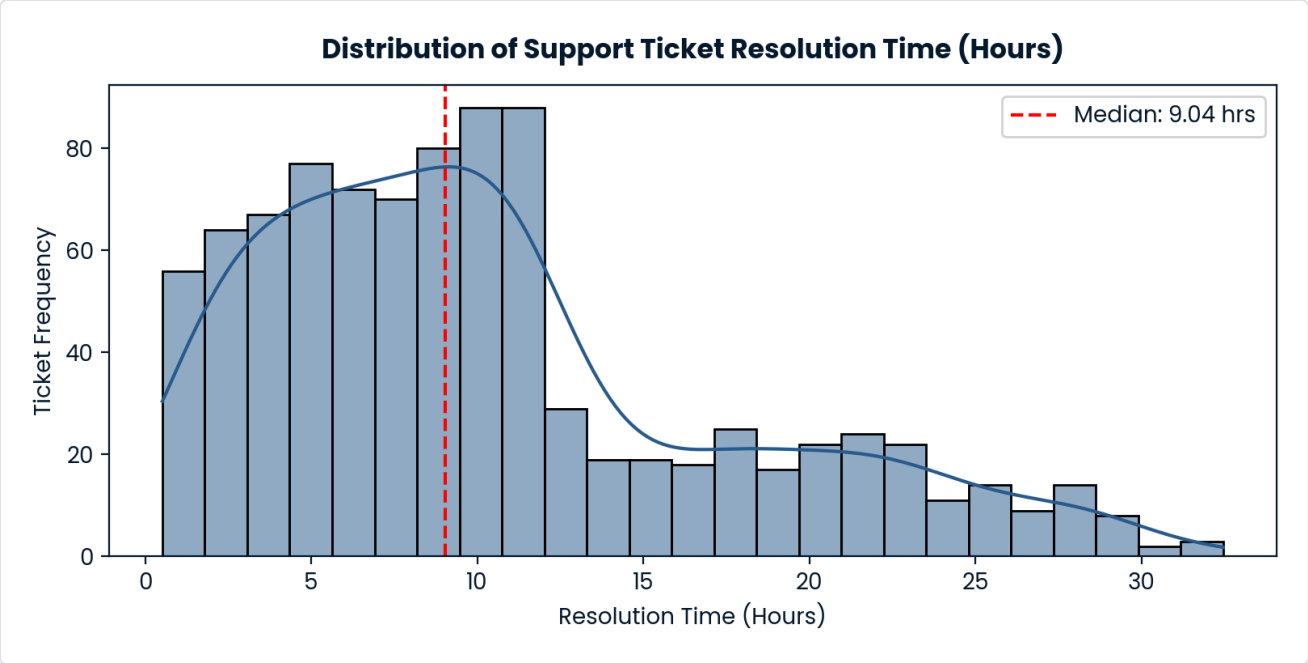


Chart Type: Continuous Histogram with Kernel Density Estimation (Single Variable)

Findings:

- Support ticket resolution time exhibits a heavily **right-skewed distribution**.
- The **median resolution time is 9.04 hours**, showing that over 50% of customer tickets are resolved within a reasonable timeframe (under 10 hours).
- A significant long tail extends from 15 to over 30 hours, indicating operational bottlenecks in handling complex or escalated support queries.

Graphic 3: Multi-Variable (Side-by-Side Boxplot - Numeric vs Categorical)

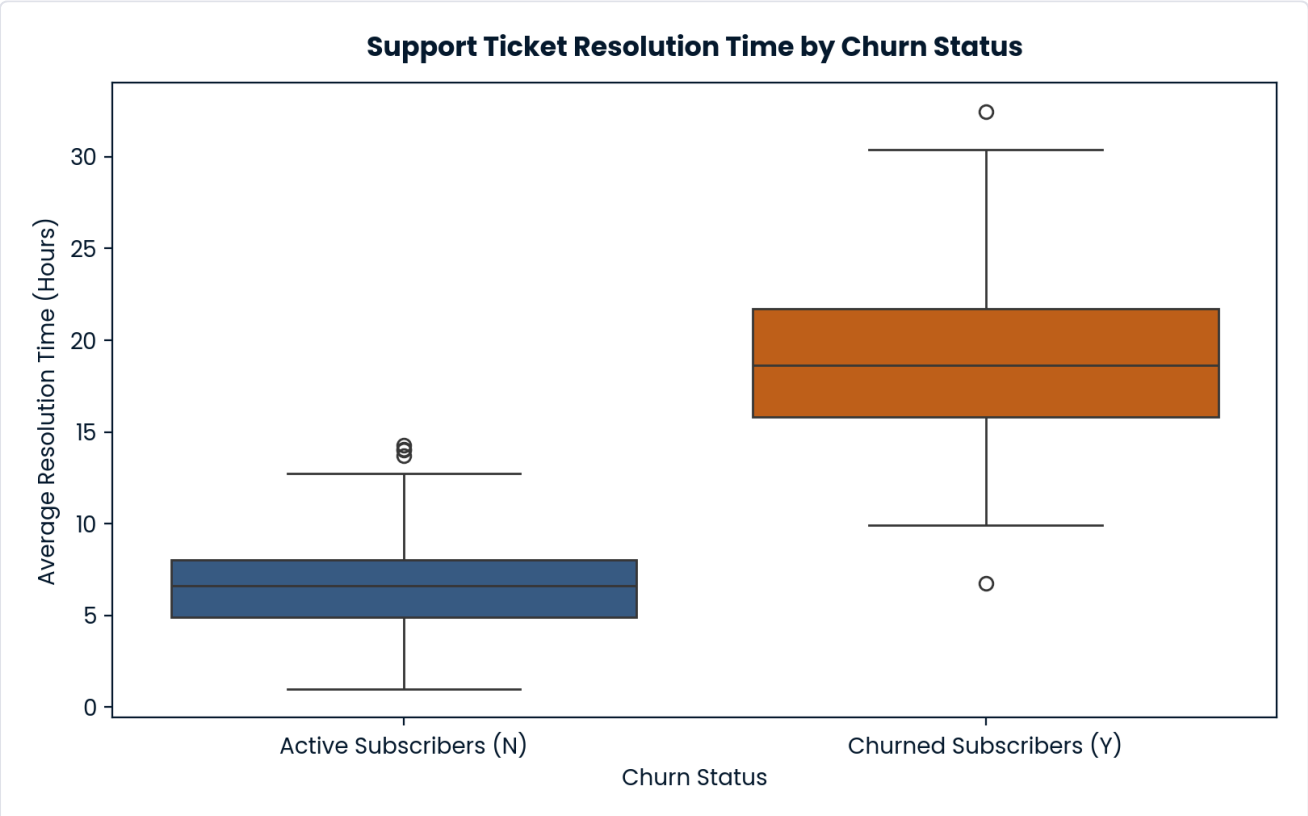
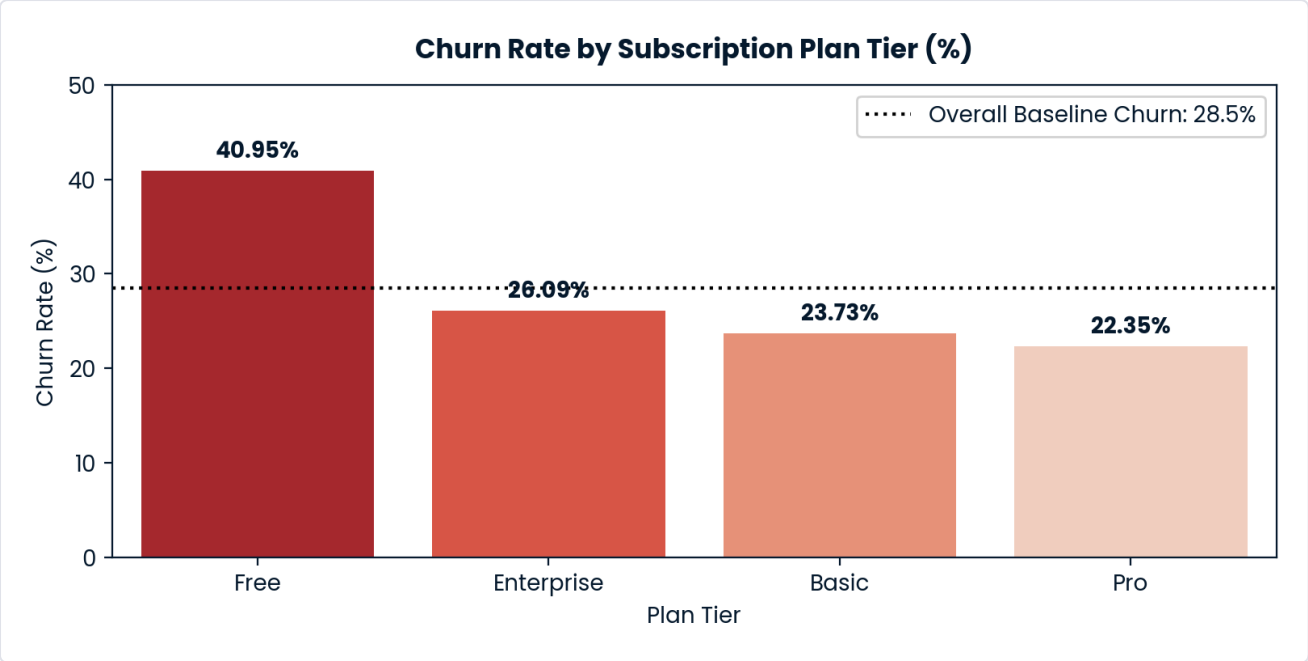


Chart Type: Boxplot (Multi-variable Comparison)

Findings:

- A clear relationship exists between support resolution delays and customer churn.
- **Active Subscribers (`N`):** Experience prompt resolution times, with a median average resolution time of approximately **6.5 hours** (IQR between 4.5 and 8 hours).
- **Churned Subscribers (`Y`):** Suffer from significantly higher wait times, with a median average resolution time of **18.5 hours** (IQR between 16 and 21.5 hours).
- **Key Takeaway:** Unresolved or slow support tickets (exceeding 15+ hours) are a primary operational driver of subscriber churn.

Graphic 4: Multi-Variable (Churn Rate by Plan Tier)



Findings: Churn Risk Disparity Across Plans

- **Free Plan Vulnerability:** Free plan subscribers exhibit a significantly higher churn rate (**40.95%**) compared to paid tiers (**Basic: 23.73%, Enterprise: 26.09%, Pro: 22.35%**).
- **Baseline Comparison:** The overall baseline churn rate across all 400 accounts stands at **28.50%**.
- **Key Takeaway:** Free tier users churn at nearly double the rate of Pro tier subscribers (**22.35%**), highlighting onboarding friction during trial or freemium usage.

Key Exploratory Findings

1. **Subscriber Base Distribution (Graphic 1):** Basic (**118**) and Free (**105**) tiers account for the majority of subscribers, while premium tiers represent smaller proportions (**Enterprise: 92, Pro: 85**)
2. **Support Ticket Latency (Graphic 2):** Support resolution time is right-skewed with a median of **9.04 hours** and a long tail extending past **30 hours,** signaling operational bottlenecks
3. **Resolution Delays Drive Churn (Graphic 3):** Active subscribers experience prompt resolution (**median 6.5 hours**), whereas churned subscribers suffer severe delays (**median 18.5 hours**)
4. **Plan-Level Churn Vulnerability (Graphic 4):** Free tier accounts face a disproportionately high churn rate (**40.95%**) compared to paid plans (**22.35%–26.09%**) and the baseline average (**28.50%**).

Definition of a metric

To reduce subscriber churn, Fit.ly should establish **Average Support Ticket Resolution Time (Hours)** as its primary operational KPI.

Operational Framework & Usage

1. **Customer Support Management (SLA Enforcement):**
    - Establish a target Service Level Agreement (SLA) of resolving **90% of tickets within 6.0 hours**.
    - Monitor resolution time daily to identify staffing gaps or complex issue categories causing resolution bottlenecks.
2. **Customer Success & Re-engagement Workflows:**
    - Automatically flag any account experiencing a support resolution time exceeding **12.0 hours** in the CRM.
    - Direct customer success representatives to perform proactive outreach (e.g., check-in emails, extended free trial periods, or subscription discounts) to high-risk accounts prior to potential cancellation.
3. **Product & Engineering Teams:**
    - Track resolution times by ticket category to pinpoint recurring technical bugs or onboarding hurdles that disproportionately impact Free and Basic plan tiers.

Initial Baseline Metric Values

Metric Indicator	Baseline Value
Median Support Ticket Resolution Time (Overall)	9.04 Hours
Active Subscriber Median Resolution Time	6.50 Hours
Churned Subscriber Median Resolution Time	18.50 Hours
Target SLA Operational Goal	< 6.00 Hours

Final summary and recomendation

Summary of Analysis

Analysis of 400 Fit.ly subscriber accounts and customer support records reveals an overall churn rate of **28.50%.** Strategic churn risk is concentrated across two main operational areas:

1. **Support Bottlenecks Drive Cancellations:** Unresolved support friction is a primary driver of customer attrition. Active subscribers experience prompt median resolution times of **6.50 hours**, whereas churned subscribers suffer severe delays averaging **18.50 hours**.
2. **High Vulnerability in Free Tier:** Free plan users churn at **40.95%**, which is nearly double the rate of paid tiers (**22.35%–26.09%**), indicating onboarding drop-off and friction when users first test the platform.

Actionable Business Recommendations

1. **Implement Priority SLA Routing for Support Tickets:**
    - Redesign ticket routing workflows to guarantee target resolution times under **6.0 hours**. Prioritize high-impact billing and onboarding tickets to prevent customer frustration.
2. **Automate High-Latency Proactive Outreach:**
    - Build automated CRM alerts whenever an open ticket exceeds **12.0 hours**. Prompt customer support leads to intervene with personalized check-ins before the subscriber initiates account cancellation.
3. **Enhance Free Tier Onboarding & Self-Service Support:**
    - Introduce interactive in-app onboarding guides and expanded self-service FAQs tailored for Free plan users during their first 14 days, reducing support ticket volume and lowering initial trial drop-off.